TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT V : TOOL DESIGN FOR CNC MACHINE TOOLS

Syllabus (as given)

Introduction - Tooling requirements for Numerical control systems - Fixture design for CNC machine tools - Sub plate and tombstone fixtures - Universal fixtures - Cutting tools - Tool holding methods - Automatic tool changers and tool positioners - Tool presetting - General explanation of the Brown and Sharp machine.

1. Introduction

- *Numerical Control (NC):* machine tool movements and functions are controlled by coded numerical instructions (program) - G-codes (motion) and M-codes (miscellaneous).
- *CNC (Computer Numerical Control):* an on-board computer (controller) reads the part program, interpolates the paths and drives the servo motors of axes with feedback from encoders/ scales.
- *DNC:* many CNC machines fed from a central computer. CNC machining centre (MC) = milling machine + ATC + pallet changer; turning centre = lathe with turret and live tools.

Feature	Conventional machine	CNC machine
Control	operator's skill and hand wheels	program + servo drive
Accuracy	operator dependent	+/- 0.005 mm, repeatable
Set-up	jigs/fixtures needed for every job	simple universal fixtures , offsets in controller
Tool change	manual	automatic (ATC)
Flexibility	low	high - change the program
Tooling cost	high (special tools)	general-purpose tooling , modular

- *Axes:* Z is along the spindle; X and Y in table plane (milling); A, B, C rotary about X, Y, Z.

Coordinate systems: machine zero (home), work offset (G54-G59) , tool length offset (G43) , cutter compensation (G41/G42) .

- Consequence for tooling: CNC aims at low non-cutting time , few set-ups , machining many faces in one clamping -> fixtures and tools must be accurate, rigid, quick-change, and presettable .

2. Tooling Requirements for Numerical Control Systems

- Tools for CNC must be:

1. More rigid and accurate - because the machine runs at high speed and feed , with unattended operation.
2. Wear resistant with long and consistent life - carbide / coated / ceramic / CBN / PCD inserts ; because tool change interrupts production (tool life is planned , tool life management in the controller).
3. Standardised and quick-change - standard shank (BT, CAT, HSK, ISO) , standard tool length and diameter ; modular tooling system.
4. Presettable offline - length and diameter measured outside the machine -> offsets are entered (no machine downtime).
5. Good chip control - chip breakers , high-pressure coolant; since the operator isn't there to clear chips; chips must not wind round the tool .
6. Reliable and fool-proof - tool monitoring (breakage detection, tool load) .
7. Dimension stability - thermal stability of holders; balanced for high rpm (G2.5 at 15 000 rpm).
8. Minimum variety of tools - use multi-purpose tools so that ATC magazine capacity (20-100) suffices.

Main differences from conventional tooling

- Tool is located by the machine using offsets -> no set block / feeler gauge setting
- Work is located by fixture + probe ; datum known to the program
- Everything in tooling is predictable : the programmer decides cut conditions , so repeatability is vital
- Tooling economics: higher cost per tool but lower per part (long life, low scrap)

- Cutting tool materials for NC: HSS-Co, coated carbide (TiN, TiCN, TiAlN, Al2O3 CVD / PVD) , cermet, ceramic, CBN, PCD ; indexable inserts dominate (CNMG, DNMG, WNMG for turning, SDMT, APMT for milling).

3. Fixture Design for CNC Machine Tools

- A CNC fixture locates the work accurately relative to machine zero , clamps it rigidly against high forces , gives full access to the tool (multi-face machining) and allows quick changeover .

3.1 Requirements specific to CNC fixtures

- Zero (datum) point: the fixture has a known and repeatable datum (the fixed corner/ locator , a zero-point system) related to the machine origin ; the programmer uses it as work offset (G54 ...) . So fixture position must be fixed accurate and repeatable .
- Positive, accurate location by dowel pins and keys in table T-slots or grid holes , or zero-point clamping system (repeatability 0.005 mm).
- Full tool access: no clamps in the tool path (clamps low profile , pulled-back), tool can approach 5 faces in one set-up; check by simulation (collision).
- Rigidity: clamps near the cutting and over supports; vibration damping ; base thickness enough ; minimum overhang ; short tool length .
- Chip evacuation: open design, slots, inclined surfaces, high-pressure coolant and chips must not accumulate on locators ; through holes in the base .
- Modularity and standardisation: standard grid plates, tooling blocks, risers, toe clamps, 'T-nuts', hydraulic vices.
- Quick changeover: hydraulic / pneumatic clamping , pallets (work is loaded outside the machine while the machine cuts the previous pallet).
- Fool-proofing and verification: part presence sensors; probe checks; locators that prevent wrong loading.
- Light weight (for pallet-changers) , the weight limit of table/pallet load .

3.2 Types of clamping used

- Mechanical: strap and toe clamps, edge clamps (pull-down), step clamps , screw-type 'side' clamps .
- Hydraulic: swing clamps, work supports (retractable), hydraulic vices, pull-down clamps; high force, fast, uniform.
- Pneumatic: quick for light cuts ; vacuum fixture: for sheets and plates (large flat area, no distortion).
- Magnetic (permanent / electro) , low-melting alloy or wax embedding for delicate shapes (turbine blades).

3.3 Design procedure (CNC fixtures)

1. Study part & process plan: operations, face to be machined, datums, tolerance, cutting forces.
2. Choose machine and table layout (T-slot pitch, hole grid, pallet size, travel).
3. Select locating scheme (3-2-1) from the finished/ machined datum ; first operation on rough faces by adjustable pads .
4. Design clamping (positions, force) so that tool path is free .
5. Decide zero point and offsets ; probe cycle for alignment if required.
6. Check rigidity (deflection < 1/3 tolerance) , tool access (collision) , chip & coolant flow .
7. Draw, simulate with CAD/CAM, prove out by the first-off and revise .

- Fixture cost guidelines: keep to 10-20 % of the part cost over the production run ; use modular where lots small .

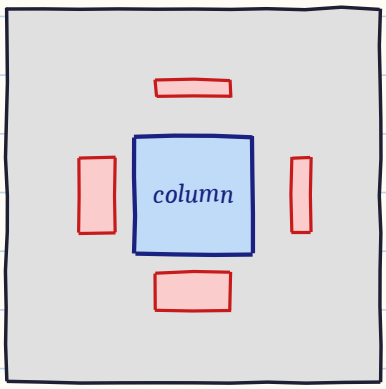
4. Sub-plate and Tombstone Fixtures

4.1 Sub-plate (grid plate) fixtures

- A sub-plate is a flat, thick, precisely ground plate (steel or cast iron) with a grid of accurately spaced tapped holes and dowel holes (e.g. 25 or 50 mm pitch, +/- 0.01 mm) that is bolted and dowelled to the machine table or pallet once and stays on the machine .
- Work-holding elements (clamps, locators, vices, risers, angle plates) are mounted anywhere on the grid with a fixed known position ; so the fixture is built quickly and repeated by recording coordinates.
- Advantages: quick set-ups, reusable, accurate and repeatable positions (known coordinates), easy to program (the position of every locator is known) , protects the machine table from damage; a standard for small-lot production .
- Types: plain grid (tapped + dowel holes), T-slot sub-plate, vacuum sub-plate, a sub-plate with a pallet system; tooling plate with a hole-pattern for zero-point clamping .
- Limitation: Limited surface, can't be used for machining all faces ; height is added -> reduces the travel .

4.2 Tombstone fixtures

Top view : 4 faces, 4 parts



Tombstone fixture

Tall block (cube / column) mounted on a pallet or rotary table of a horizontal machining centre (HMC).
Parts clamped on 2, 3 or 4 faces.
Table indexes 90 deg -> next face cuts without stopping -> less idle time.
Used with pallet changer: load one tombstone while the other is machining.

- A tombstone is a tall rectangular (or square) column with work-holding on its 2, 3 or 4 vertical sides , fixed on a pallet or the rotary table of a horizontal machining centre (HMC) .
- Operation: machine the part on one face , index the table (B-axis) 90 deg (or 180), machine the next face... each face can hold one or many (many identical parts, or several different parts, 'family' of parts) ; the operator loads the faces on the other pallet while the machine is cutting .
- Advantages: high spindle utilisation (low idle time), more parts per set-up, more tools used on the single loading, multiple faces by one fixture, accurate relationships between faces (single set-up), fewer pallets for pallet pool , good chip fall (vertical faces) .
- Design points: tombstone is heavy and rigid ; keep the centre of gravity low and over the table centre ; height within the Y-travel and tool reach ; chip fall-through gaps ; strong (grid / dowel hole) faces ; face-to-face accuracy : faces ground parallel/ perpendicular within 0.01 mm ; coolant and hydraulics through-the pallet for the hydraulic clamps (rotary union on the table) .
- Variants: cube (4 faces), 'dual-face' (2 sides), 'triangular' ; cast iron or welded steel/ aluminium , with a grid or T-slots on each face.

4.3 Comparison

Item	Sub-plate (grid plate)	Tombstone
Position on machine	flat on table / pallet	vertical column on pallet (HMC)
Faces machined	1 (top) + sides with angle plate	2 to 4 sides, many parts

Item	Sub-plate (grid plate)	Tombstone
Use	VMC / small lot, many different jobs	HMC, medium to high volume
Set-up time	short	very short per part (many parts/set-up)
Main advantage	flexibility, known grid	spindle utilisation, accuracy of faces

5. Universal Fixtures

- Universal (general-purpose / flexible) fixtures: standard work-holding that can be used for a wide range of different parts with little or no modification - the fixture is not designed for one part .
- Examples: machine vices (with soft jaws), chucks (3-jaw, 4-jaw, collet, magnetic), rotary tables and index heads, angle plates, V-blocks, parallels, step blocks and clamps, swivel / tilting tables (trunnion), vacuum tables, tooling-plate kits, modular fixturing systems.
- Modular fixture systems: a set of standard hardened and ground elements (base plates with T-slot or hole grid, riser blocks, locators, supports, clamps) are assembled into a fixture for a job, used, then dismantled and kept for reuse .
  - Types: T-slot based modular system dowel-pin / hole-based system (grid; '16 mm hole, 50 mm pitch' common), and Combi-/ phase systems.
  - Advantages: low lead time (hours), reuse of components, lower cost for small lots, accurate (precision elements), good for prototypes, flexible manufacturing systems (FMS) .
  - Limitations: less rigid (more joints), initial cost, assembly skill needed, and bigger size than a dedicated fixture .
- Other universal approaches: hydraulic / zero-point clamping on all vices and plates , adjustable (reconfigurable) pin-bed / 'bed-of-nails' fixtures , phase-change (low-melting alloy, ice) embedding , and programmable / adaptive fixtures with robot .
- Group technology (GT): the parts of a family of similar shape are held in one adjustable fixture (' group fixtures '), by changing the inserts, jaws or locators only.

Fixture type vs production volume

- Low volume, many variants -> universal / modular / sub-plate
- Medium volume, families -> tombstone, group (adaptable) fixtures
- High volume, single part -> dedicated hydraulic fixture / pallet

6. Cutting Tools for CNC

- Types of cutting tools used on machining centres: face mills, end mills (solid carbide / indexable), ball-nose and bull-nose cutters, slot drills, shell mills, T-slot and thread mills, drills (twist, indexable, through-coolant, spade), reamers, boring heads (fine adjustable), taps (rigid tapping), chamfer and counter-sink tools, form tools .
- Turning centre tools: ISO (indexable) turning inserts and holders (external, internal, grooving, parting off, threading), boring bars (damped), drills , live (driven) tools for milling on the lathe.

6.1 Indexable insert tools

- Inserts clamped on the tool body by screw, top-clamp, lever, wedge or clamp-pin ; edge indexed to a fresh edge when worn - no regrinding; consistent size (+/- 0.01) . Several shapes (C, D, R, S, T, V, W) , clearance angles (N, B, C ..) , tolerance classes (G, M, U) , chip-breaker geometry (Seen in ISO 1832 code : e.g. CNMG 120408 = C shape 80 deg, 0 deg clearance (N), tolerance M, hole with chip-breaker (G), size 12 mm, thickness 04, nose radius 0.8 mm).
- Selection factors: work material (ISO P - steel (blue), M - stainless (yellow), K - cast iron (red), N - non-ferrous (green), S - superalloy (brown), H - hardened (grey)) ; operation (roughing, finishing) ; machine power & rigidity ; tool life & cost .
- Tool geometry for NC: generally positive-rake inserts for finishing and low power , negative-rake (strong, double-sided so with 2 x the edges) for heavy roughing .

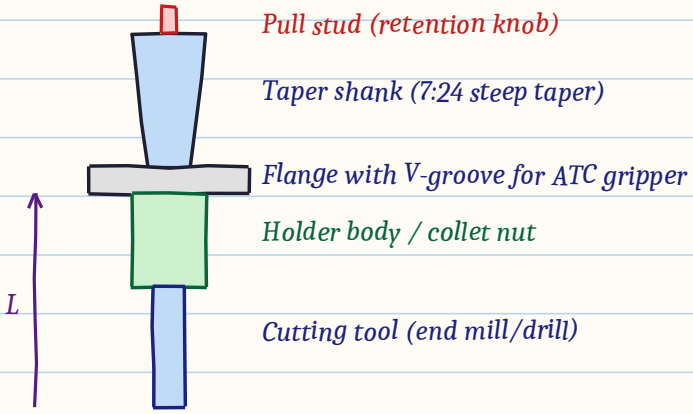
6.2 Tool selection rules for CNC

1. Choose largest suitable tool diameter and the shortest possible stick-out (overhang) for rigidity (deflection ~ L3/D4).
2. Use standard tool sizes and 'one tool, many jobs' (minimising the tool changes).
3. Use through-spindle coolant tools for deep holes and chip evacuation.
4. For finishing use wiper inserts and light cut ; for roughing use coarse pitch cutters with chip splitters .
5. Tool life management: life in minutes or parts counted by controller, sister tools

(duplicate) used automatically when life is over.

6. Spindle speed N = 1000 V/(pi D) ; feed rate F = fz x z x N ; cutting data from catalogue .

7. Tool Holding Methods



Tool presetting data

Tool length L (gauge line to tip)
Tool diameter D (radius offset)
Measured OFF the machine on a presetter -> values entered in the tool offset table of the CNC

- Tool holder: a standard interface between the machine spindle and the cutting tool ; it must provide accurate location (concentric, low runout < 0.005 - 0.01 mm), high clamping force, high stiffness, balance, and be compatible with the ATC (gripper groove + pull stud) .

7.1 Spindle (shank) interfaces

- 7:24 steep taper (self-releasing; ISO / BT / CAT / DIN 69871 (SK) 30, 40, 50) - most common on machining centres , the taper is pulled into the spindle by a drawbar and pull-stud (retention knob) ; flange is gripped by ATC.
- BT (Japanese; symmetric - good for high speed), CAT (American V-flange), SK/DIN (German), ISO.
- HSK (hollow short taper 1:10): hollow taper shank + face contact (dual contact) -> high accuracy/ stiffness at high rpm (above 20 000 rpm) e.g. HSK-A63; Capto (polygon taper PSC) for turning/ milling.
- Morse taper (MT 1 - 6), R8, 'Weldon' flats on older / drilling machines.
- Turning: VDI (DIN 69880) and BMT (base mount) turret tool holders , square shank (25 x 25) holders , boring bars clamped in bushes , quick-change tool posts .

7.3 Methods of holding the cutting tool in the holder

Method	How	Remarks

Method	How	Remarks
Collet chuck (ER collet)	nut closes a slotted collet around the shank	versatile, runout 0.01 mm , common; many sizes
Side-lock (Weldon) holder	screw presses on a flat on the tool	high torque, no pull-out; poor concentricity - roughing
Shrink-fit holder	holder bore heated (induction 300 C), tool put in; shrinks on cooling	run out < 0.003 mm, perfect balance, high stiffness , high rpm; need heater
Hydraulic chuck	hydraulic pressure expands a thin sleeve	quick, vibration damping, 0.003 mm accuracy
Milling chuck (high-torque)	needle roller bearings force a sleeve	high grip for heavy roughing
Face-mill arbor (shell mill holder)	tool bolted on the face, driven by a key	big face mills
Drill chuck / Jacobs, tapping holder	keyless drill chuck; compression-tension tapping head	drilling, tapping
Boring head	micro adjustable slide	bore sizes with 0.002 mm adj.
Modular / quick-change	tool head screwed or taper-locked onto the shank	reduces tool inventory

- Balancing: unbalanced holder -> vibration at high rpm; G2.5 or G6.3 balancing grade ; symmetric design , nut with balance screws .

8. Automatic Tool Changers and Tool Positioners

8.1 Automatic tool changer (ATC)

- ATC swaps the tool in the spindle for the next one automatically on the program command (M06, T-code) within 1 - 10 seconds (chip-to-chip 2-6 s) , which makes the multi-operation machining possible without an operator .
- Main parts: tool magazine (storage) , tool changer arm (gripper) / mechanism , tool-selection (positioning) system , spindle with drawbar and clamp (spring clamp + hydraulic unclamp) .
- Types of magazines:

1. Drum (disc) magazine: tools in the radial / axial pockets of a rotating disc; capacity 12 - 40 ; the whole magazine moves to spindle (shuttle) ; simple, fast .
2. Chain magazine: tools on a chain with sprockets ; capacity 40 - 120+ ; (horizontal, vertical or oval) ; big and flexible .
3. Rack (matrix) / bin magazine: tools in rows , accessible by robot / gantry ; capacity 100 - 500 ; used in large machines & FMS.
4. Turret (lathe) : 8 - 12 stations indexed (servo or Hirth coupling), the tool changing = turret indexing in 0.2 - 1 s .

- Changing methods: (a) Arm type (double-arm, 180 deg swing) : gripper grabs the new and old tools simultaneous - fastest; (b) Direct spindle-to-magazine ('pick-up') : spindle moves to the magazine; no arm - cheaper, slower; (c) Robot / gantry type .
- Tool selection: Sequential (tools in the order of use; simple), random (memory) selection - the controller keeps track of the tool's pocket; the tool is put wherever the pocket is free; 'fixed' pocket addressing with tool code (bar code, RFID chip).

ATC sequence (random / arm type)

- Magazine indexes the next tool to standby position (parallel to cutting)
- On M06: spindle goes to the tool change position (Z home), orients (M19)
- Arm swings: grips both tools -> unclamps spindle -> pulls out -> rotates 180 deg -> inserts -> clamps
- Arm returns; spindle goes back and magazine places the old tool in the pocket

8.2 Tool positioners

- Tool positioner: a device that accurately brings the tool into a defined (known) position or indexes tool holders , e.g. the lathe turret , the tool-setter (probe on machine) , a tool pre-setter or the magazine indexing mechanism .
- Features: Hirth (serrated) coupling in turret for repeatability 0.002 mm; Geneva mechanism / servo with encoder for magazine indexing ; keys for tool orientation (the drive keys on the flange, spindle orientation M19) .
- Tool offsets (length and radius) compensate for the differences in tool position (so the program uses nominal values).
- Tool monitoring / tool breakage detection : touch probe or laser, spindle-load monitoring, acoustic emission , tool-life counters .

9. Tool Presetting

- Tool presetting = measuring the length and diameter of each tool (with its holder) outside the machine in a tool presetting device , and entering the values as tool offset in the controller, so the machine is not stopped for tool setting .
- Presetter: a precision spindle with the same taper (e.g. BT40) mounted on a column , with optical (shadow-graph/ camera) / touch-probe measuring unit and a digital display ; data transfer by RFID chip / data cable / DNC to the machine .
- What is measured: tool length L from the gauge line (flange) to the cutting tip, the radius/diameter D, the nose radius, run-out, and profile check .

Benefits of presetting

- Machine idle time is reduced (no on-machine setting)
- Tool life management and consistent first-part accuracy
- Detects tool defects, wrong tool or runout before use
- Preset data stored with tool ID (RFID), avoid human errors in offset entry

- Methods: (1) off-line presetting (optical / contact) , (2) on-machine tool setters (touch probe / laser) - measure automatically during the program and update wear offset , (3) manual touching of reference block / 'touch-off' (slow but no equipment) .
- Offsets used: G43 H.. = tool length offset ; G41/G42 D.. = cutter radius compensation ; wear offsets are modified after first part inspection .
- Typical accuracy of presetters: length +/- 0.005 mm; diameter +/- 0.003 mm ; temperature controlled room and calibrated masters .

10. General Explanation of the Brown and Sharp Machine

- Brown & Sharpe is a famous American machine-tool and precision-instrument maker (David Brown, joined by Lucian Sharpe; the name Brown & Sharpe dates from 1853). It is known for the universal milling machine (1862), the micrometer (1867), grinders, measuring machines (later CMMs) and the automatic screw machine .
- In this unit the Brown & Sharpe machine is explained as the classic cam-controlled automatic screw machine : bar stock is fed through a hollow spindle, held by a collet, and cut by tools on cross slides and a turret, all movements controlled by cams - the mechanical forerunner of CNC for mass producing small parts (screws, pins, bushes).
- (Exam note: also quote your prescribed textbook's description of the Brown & Sharpe machine and its control system if your faculty has stressed it.)

10.1 Main parts

- Headstock with hollow spindle and collet chuck (feed and clamp by cams) , bar stock magazine and feed finger .
- Cross slides (front, rear, vertical) carrying form tools, cut-off tool ; turret (hexagonal, horizontal axis or sliding) with 5-6 stations for drills, reamers, dies, taps, box-tools.
- Camshaft (main cam shaft) with cam drums / plates which control (a) the slides, (b) the turret feed and index, (c) the collet opening & closing and bar feed, (d) speed change ; gear train for speeds and feeds.
- Stock stops, work holding on collet, coolant system.

10.2 Working and tooling

- Working cycle (one revolution of the cam shaft = one component): feed bar to the stop -> close the collet -> turret indexes to station 1 -> form tool/ drill/ turn/ thread -> cut off -> open collet . Cycle time set by cam speed change gear .
- Tooling: form tools (circular / dovetail) on the cross slide for contours , box tool for turning on turret, knee tools , drills & reamers , taps & dies (self-opening die heads), knurls , cut-off blade .
- Set-up: cam design (lobes with rise and fall give feed rate) , tool layout chart (sequence of tools at each station and cam) , form tool design (relief angle 10-15 deg & rake 0-15 deg).
- Advantages: full automatic (one operator for many machines) , very high production , good accuracy for small components ; tooling is simple. Limitations: long set-up time (cams), limited to bar stock, not flexible; replaced by CNC 'Swiss-type' lathes and turning centres.
- Relevance to NC: CNC replaces mechanical cams by program and servo drives - this removes the cam changing and gives flexibility ; the same principles of tool layout, tool holding, and tool-life management still apply .

11. Quick Revision - Unit V

- NC tooling: rigid, quick-change, presettable, long life, good chip control; standard shanks (BT, CAT, HSK).
- CNC fixtures: known zero point, full access, rigid, quick change; sub-plate = grid plate; tombstone = vertical multi-face on HMC; universal / modular = reusable standard elements .

- Tool holding: 7:24 taper + pull-stud, HSK; collet (ER), shrink-fit, hydraulic, side-lock.
- ATC: magazine (drum, chain, rack) + arm; random selection with memory; turret on lathes.
- Tool presetting: measure L and D offline -> offsets G43 / G41; Brown & Sharp: cam-operated automatic screw machine (pre-CNC automation).